CONTROLADOR DE VELOCIDAD LAZO CERRADO TDA1085C – AGUJEREADORA DE BANCO (ACOPLE DIRECTO)
Motor C.E.SET. MCA 52/64–148/ALL7 (Medido: R_estator[P1–P5]=2.1R + R_rotor[P2–P6]=6.5R = 8.6R, L=25–30mH, Tacho[P3–P4]=73.0R 8P 166.6Hz@2500RPM)
Bornera 7P (Izq–>Der): P1(Gris osc: Campo A) puenteado con P2(Violeta: Carbon Izq) | MOTOR_OUT(J1): P5(Negro: Campo B)–>L2_AC_HOT y P6(Azul: Carbon Der)–>MT2 | P7(Gris claro: Sin uso)
Triac: BTB12–600BW (I_GT=50mA, ALETA VIVA NO AISLADA CONECTADA A MT2 – AISLAR DEL CHASIS)
Rango: 400 a 2.500 RPM (Preset 50k multivuelta RV1 en serie con Potenciometro 10k de panel)
Star Ground: L1 entra a MT1 de Q1 y sale en pista unica hacia Pin 8 (GND_CHIP)

J1
GND_CHIP 1 L1_MT1
MT2 2 MT2_MOT_PIN6
L2_AC_HOT 3 L2_MOT_PIN5
L2_AC_HOT 4 L2_AC_HOT
AC_220V_a_MOTOR

Q1
GND_CHIP 1 MT1
2 MT2_TAB
3 GATE
MT2
GATE
BTB12–600BW (NO AISLADO)

R_SNUB1
MT2 — N_SNUB
100R 1W
C_SNUB1
N_SNUB — GND_CHIP
100nF 400V X2

R_GATE1
PIN13_TRIG — GATE
150R 1W

D1
L2_AC_HOT — N_RECT
1N4007

R_DROP1
N_RECT — N_DROP_MID
4.7k 3W (Elevar 5mm)

R_DROP2
N_DROP_MID — VCC
4.7k 3W (Elevar 5mm)

C_VCC1
VCC — GND_CHIP
220uF / 25V

C_DEC1
VCC — GND_CHIP
100nF

R1
MT2 — PIN1_ISYNC
820kR1/2W

R2
L2_AC_HOT — PIN2_VSYNC
820kR1/2W

R3
PIN3_ILIM — GND_CHIP
10k

C1
PIN3_ILIM — GND_CHIP
100nF

RV1
VCC 1
2 N_POT_TOP
3 N_POT_TOP
50k Multivuelta

J2
N_POT_TOP 1 TOP
PIN5_VSET 2 WIPER
N_POT_BOT 3 BOT
POT_PANEL_10k

R_MIN1
N_POT_BOT — GND_CHIP
2.2k (Piso 2400 RPM)

C2
PIN5_VSET — GND_CHIP
100nF

J3
TACHO_AC 1 Pin_1
GND_CHIP 2 Pin_2
TACHO_P3_P4 (73.0R 8P)

R12
TACHO_AC — PIN12_TACHO
10R

C12
PIN12_TACHO — GND_CHIP
10nF (R11 Picos)

R11
PIN11_PUMP — GND_CHIP
470R 1nF

R10
VCC — PIN11_PUMP
47k

R9
PIN4_VTACHO — GND_CHIP
27k

C11
PIN4_VTACHO — GND_CHIP
470nF

C7
PIN7_RAMP — GND_CHIP
4.7uF / 50V

C14
PIN14_SAW — GND_CHIP
47nF (1KHz Saw)

R15
VCC — PIN15_ISAW
27k

U1 TDA1085C
PIN1_ISYNC 1 I_SYNC
PIN2_VSYNC 2 V_SYNC
PIN3_ILIM 3 I_LIM
PIN4_VTACHO 4 V_TACHO
PIN5_VSET 5 V_SET
GND_CHIP 6 GND_CHIP
PIN7_RAMP 7 RAMP
GND_CHIP 8 GND_CHIP
16 LOOP_COMP PIN16_COMP
15 SAW_CURR PIN15_ISAW
14 SAW_CAP PIN14_SAW
13 TRIG_OUT PIN13_TRIG
12 TACHO_IN PIN12_TACHO
11 TACHO_PUMP PIN11_PUMP
10 VCC
9 +VCC VCC

R16
PIN16_COMP — N_COMP_INT
47k

C16
N_COMP_INT — GND_CHIP
1uFC1750V

C15
PIN16_COMP — GND_CHIP
47nF

Triac BTB12–600BW (No Aislado, I_GT=50mA, R_G=150R 1W) | DRC: Clearance >3.0mm | Star Ground
Serie: Puente P1(Gris osc)–P2(Violeta) | MOTOR_OUT: P5(Negro)–>L2_AC_HOT, P6(Azul)–>MT2
Motor C.E.SET. MCA 52/64: R_est(P1–P5)=2.1R + R_rot(P2–P6)=6.5R = 8.6R | Tacho(P3–P4)=73.0R
**UTN – Hardware Libre**

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***Title: Controlador TDA1085C para Agujereadora de Banco (Acople Directo)***

| Size: A3 | Date: 2026–11–11 | **Rev: 2.1** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |